

# Rongfeng Cheng | Medical AI Research

Guangzhou, China

✉ eb1287945452@mail.scut.edu.cn • 🌐 crf2004.github.io/en • 🏠 CRF2004  
0009-0002-3021-9647

## Honors & Awards

**2025:** Gold Award, Internet+ University Competition – Dingbei Xinxin cardiovascular health-management project.

**2025:** Second Prize, 9th Guangzhou Mathematical Modeling University League – led modeling and algorithm implementation for a ConvNeXt-based fruit-image classification pipeline with task-specific architecture adaptation and data augmentation.

**2025:** Third Prize, China Undergraduate Mathematical Contest in Modeling, Guangdong Division – led modeling and algorithm implementation for smoke-screen deployment using nonlinear programming, genetic algorithms, and particle swarm optimization.

## Education

### South China University of Technology (SCUT)

Guangzhou, China

*Bachelor of Management in Big Data Management and Application*

2023–2027

GPA: 3.46/4.0.

Selected AI coursework: Large Models and Generative AI (93/100), Large Language Models and Prompt Engineering (93/100), Machine Learning (92/100), and Deep Learning (88/100).

## Research & Project Experience

### Institute of Healthcare Artificial Intelligence Application, Guangdong Second Provincial General Hospital

*DLEF: Dataset-Label Evidence Fingerprints*

Jun 2026–Present

Advisors: Haobin Zhang and Weibin Cheng.

- Solely led study design, implementation, experiments, analysis, and manuscript preparation; proposed directional evidence fingerprints for characterizing dataset-label transfer risk.
- Across CODE-15%, PTB-XL, and Chapman-Shaoxing, DLEF distance was associated with label-level external AUROC degradation (Spearman  $r = 0.843$  across 108 points;  $r = 0.892$  after source-target-label aggregation). See Manuscript [1].

### Institute of Healthcare Artificial Intelligence Application, Guangdong Second Provincial General Hospital

*Evidence-Use Auditing for Deep ECG Models*

Jan 2026–Present

Advisors: Haobin Zhang and Weibin Cheng.

- Solely led study design, implementation, experiments, analysis, and manuscript preparation; developed a matched-control perturbation audit spanning clinically defined evidence units, operator comparisons, and frozen-weight external transfer.
- Evaluated five diagnoses and three raw-signal models on CODE-15% and PTB-XL; the audit classified 6 of 15 model-diagnosis pairs as evidence-supported and showed that matched controls or operator choice can overturn target-only conclusions. See Manuscript [2].

### Co-founder & Core Developer; Board Director since Aug 2026

*Dingbei Xinxin / GradioGPT: Proactive Cardiovascular Health Management LLM*

2025–Present

- Co-developed a cardiovascular health-management platform integrating PPG/ECG data, intelligent consultation, risk assessment, and proactive health workflows.
- Led model fine-tuning, RAG, and full-stack implementation using LoRA, DPO, MQ-RAG, and chain-of-thought methods; also prepared grant and competition application materials.
- Deployed the initial mini-program at Guangdong Second Provincial General Hospital and currently leads full-stack development of version 2.0; supported by national-level innovation-program projects in 2025 and 2026 and incorporated as Yujian Future Mind Technology (Zhongshan) Co., Ltd. in Aug 2026.

### Project Lead, SCUT Student Research Program

Guangzhou, China

*Myocardial Ischemia Knowledge Graph*

2025

Advisor: Xunde Dong. Led a five-student interdisciplinary team; designed the ontology, annotation schema, guidelines, and quality-control workflow; built a multi-annotator platform with adjudication and agreement checks; and implemented XGBoost-based implicit reasoning with graph-mapped interpretability analysis.

### Independent Research

*IHD Trajectory Similarity Retrieval*

Nov–Dec 2025

Developed trajectory retrieval and post-cutoff validation for ischemic heart disease ICU stays in MIMIC-IV using time-windowed clinical variables and precomputed k-NN similarity graphs.

## Manuscripts & Publications

**Under Review [1]:** R. Cheng et al. “DLEF: Dataset-Label Evidence Fingerprints for Explaining Cross-Dataset Performance in ECG Classification.” *IEEE BIBM 2026*. First author.

**Under Review [2]:** R. Cheng et al. “A Matched-Control Perturbation Audit Framework for Evaluating Evidence Use in

Deep Learning Models: An Application to 12-Lead ECG Classification.” *IEEE JBHI*. First author.

**Ongoing: PathCLIP: Evidence-Aware ECG–Text Representation Learning.** Ongoing first-author study of ECG–text alignment and zero-shot recognition with matched-count, permuted-hierarchy, residual-only, and held-out-label controls.

**Apr 2026 [3]:** J. He, S. Qin, H. Zou, F. Jing, W. Liu, C. Li, Q. Wang, Z. Hu, K. Song, Z. Xu, Q. Luo, Y. Cheng, **R. Cheng**, H. Zhang, and W. Cheng. “CodeSafetyBench: Evaluating and Improving Ethical Safety in Large Language Model Code Generation.” *iScience*. doi:10.1016/j.isci.2026.115073.

**Mar 2026 [4]:** R. Wang, F. Jing, **R. Cheng**, Y. Cheng, W. Cheng, H. Zou, H. Ren, and W. Jin. “Artificial Intelligence Applications in X-ray Coronary Angiography: A Comprehensive Review of Image Analysis Techniques.” *HNNDL* 2026. doi:10.1145/3795892.3795917.

**Mar 2026 [5]:** Y. Fang, F. Jing, **R. Cheng**, Y. Cheng, W. Cheng, H. Ren, W. Cai, and W. Jin. “Artificial Intelligence for Echocardiographic Assessment of Pulmonary Hypertension and Right Heart Function: Advances in Intelligent Diagnosis and Risk Stratification.” *HNNDL* 2026. doi:10.1145/3795892.3795918.

## Internship Experience

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### Guangdong Second Provincial General Hospital AI Research Institute

*LLM Development Engineer Intern*

*Nov 2024–Present*

- Solely developed a standardized-patient Q&A pipeline from approximately 8,000 cardiovascular case PDFs, covering LLM-based information extraction, entity disambiguation, Neo4j knowledge representation, and hybrid retrieval.
- Explored knowledge-graph paths as structural scaffolds for generating medical question–answer and reasoning-trace data for model fine-tuning, evaluation, and retrieval-augmented generation.
- Led full-stack and clinical-workflow development for the five-person Dingbei Meimei acne and facial-health assessment team; supported its initial release when the hospital’s Refractory Acne Diagnosis and Treatment Center opened in July 2026.

### Guangzhou Yunzhan Zhichuang Information Technology Co., Ltd.

**Guangzhou, China**

*Backend & Algorithm Development Intern*

*Jul 2025–Jan 2026*

- Designed a UMLS-grounded schema and normalization pipeline for heterogeneous health-examination data extracted from multi-institutional reports, including LLM-assisted schema evolution and terminology mapping.
- Developed backend workflow logic for the Health Navigation Plan within the company’s iTwins product.

## Patents & Software IP

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**Patent: Method, System, and Electronic Device for Medical Question-Answering Output Assurance** – First Inventor; application #202610582199.6; filed Apr 2026; preliminary examination passed.

**Software IP:** Knowledge Graph Ontology Intelligent Management System V1.0 (#2025SR2454864); LLM-Based Cardiovascular Proactive Health Management Intelligent System V1.0 (#2025SR2080673).

## Skills

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**Modeling:** PyTorch, XGBoost, representation learning, multimodal learning, LLM fine-tuning, optimization

**LLM & Knowledge Systems:** LoRA, DPO, RAG/MQ-RAG, knowledge graphs, information extraction, UMLS

**Evaluation & Data Curation:** RAG/retrieval evaluation, perturbation analysis, external validation, schema design, data harmonization

**Engineering:** Python, TypeScript, SQL, Neo4j, Next.js, Flask, Git, Docker

## Languages

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Chinese (native)   Cantonese (native)   English: IELTS 6.5, CET-6 528